

Quiz #4

1. A laboratory repeatedly measures a reference mass whose exact value is known to be 50.00 g. Five measurements are

50.08, 50.07, 50.09, 50.08, 50.08 g.

- (a) Are these measurements precise? Briefly explain.

Yes. The measurements are **precise** because they are tightly clustered between 50.07 g and 50.09 g. Thus, the chance-error spread is small.

- (b) Do the measurements appear to be accurate? What type of measurement error is suggested by these readings?

The measurements do not appear to be accurate. They are consistently about 0.08 g above the known value of 50.00 g. This suggests a positive **bias** (systematic error).

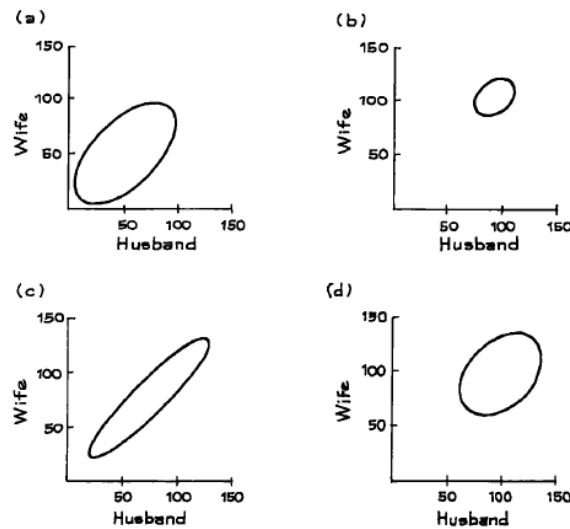
- (c) Would taking many more measurements and averaging them eliminate this problem? Briefly explain.

No. Taking more measurements and averaging them can reduce the effect of **chance error**, but it does not remove **bias**. The instrument or measurement procedure would need to be calibrated or corrected.

2. A study of the IQs of husbands and wives obtained the following results:

$$\bar{x} = 100 \quad (\text{husbands}), \quad \bar{y} = 100 \quad (\text{wives}),$$

$$s_x = 15, \quad s_y = 15, \quad r = 0.6.$$



Which scatter diagram matches these data? Briefly explain why you reject the others.

The best match is **(d)**.

The center of the scatterplot should be near

$$(\bar{x}, \bar{y}) = (100, 100).$$

Because $s_x = s_y = 15$, the horizontal and vertical spreads should be about the same. Also, $r = 0.6$ indicates a **moderate positive linear association**, so the cloud should slope upward but should not be extremely narrow.

Diagram (a) is centered too low. Diagram (b) has too little spread. Diagram (c) shows a much stronger positive association than $r = 0.6$. Diagram (d) has the appropriate center, similar horizontal and vertical spread, and a moderate positive association.

3. Suppose `gap_2007` contains the Gapminder observations from 2007. Consider the following code:

```
result = gap_2007.loc[
    (gap_2007["continent"] == "Americas")
    & (gap_2007["pop"] > 10_000_000),
```

```
["country", "lifeExp", "gdpPercap"],  
].sort_values("lifeExp", ascending=False)
```

(a) In words, which rows are selected?

The code selects countries that are in the **Americas** and have populations greater than 10,000,000.

(b) Which columns appear in **result**, and how are the rows ordered?

The resulting **DataFrame** contains the columns **country**, **lifeExp**, and **gdpPercap**. The rows are sorted by **lifeExp** from highest to lowest because **ascending=False**.

(c) Why is **&** used instead of **and** when combining the two conditions?

Each condition produces a pandas Series of Boolean values, one for each row. The operator **&** combines these Boolean Series element by element. Python's **and** is used for combining individual Boolean values, not pandas Series.